

PORTAGE LA PRAIRIE SOUTHPORT, MB, Canada

WMO: 718510

Lat: 49.9031N

Lon: 98.2839W

Elev: 895

StdP: 14.23

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-17.5	-29.4	1.1	-21.4	-24.8	1.4	-17.0	29.6	5.7	26.4	5.5	6.7	240	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	86.1	70.6	83.0	68.9	80.1	66.7	74.0	82.1	71.5	79.5	69.3	77.2	9.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	120.4	79.4	68.8	109.3	76.3	66.4	100.5	73.8	38.4	82.5	36.0	79.3	34.0	77.2	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.4	22.0	19.2	DB	-28.6	92.6	5.3	3.2	-32.4	94.8	-35.6	96.7	-38.6	98.5	-42.4	100.8	(4)
			WB	-28.6	79.2	5.4	2.7	-32.5	81.1	-35.7	82.6	-38.7	84.1	-42.7	86.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	5.3	9.1	22.5	39.2	52.1	63.0	67.9	65.9	56.6	42.6	25.7	11.2	(6)
(7)		DBStd	24.58	14.68	13.60	13.81	11.13	8.92	6.56	5.59	5.93	7.91	8.74	11.98	13.80	(7)
(8)		HDD50	6041	1386	1146	854	349	83	2	0	0	27	262	728	1204	(8)
(9)		HDD65	9934	1851	1566	1317	774	410	112	34	61	268	694	1178	1669	(9)
(10)		CDD50	1874	0	0	1	25	147	391	556	494	226	33	1	0	(10)
(11)		CDD65	293	0	0	0	0	9	51	125	90	17	1	0	0	(11)
(12)		CDH74	2643	0	0	0	6	146	466	1004	815	192	14	0	0	(12)
(13)		CDH80	656	0	0	0	0	32	104	252	227	39	2	0	0	(13)
(14)	Wind	WSAvg	8.8	9.1	9.2	9.3	10.0	9.9	8.1	7.3	7.4	8.2	9.2	9.2	8.8	(14)
(15)	Precipitation	PrecAvg	20.1	0.7	0.6	1.1	1.3	2.7	3.3	2.8	2.2	1.7	1.6	0.9	0.9	(15)
(16)		PrecMax	28.6	1.8	2.2	2.8	3.2	6.2	5.6	6.5	4.9	3.9	4.2	3.9	1.9	(16)
(17)		PrecMin	13.5	0.3	0.0	0.2	0.4	0.7	1.6	0.6	0.5	0.2	0.0	0.0	0.1	(17)
(18)		PrecStd	3.4	0.4	0.6	0.7	0.7	1.3	1.2	1.6	1.1	0.9	1.0	0.8	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	42.0	59.4	74.1	84.6	88.2	89.8	90.1	84.6	75.2	59.0	42.7	(19)
(20)		0.4%	MCWB	36.8	36.6	48.9	55.2	63.8	71.9	74.4	72.4	67.8	59.4	48.0	37.0	(20)
(21)		2%	DB	36.5	37.0	50.3	67.7	78.0	82.8	85.7	85.7	79.6	66.6	51.3	38.0	(21)
(22)		2%	MCWB	33.4	34.1	42.8	51.9	58.4	66.9	72.0	70.4	65.0	54.2	43.7	34.3	(22)
(23)		5%	DB	32.1	33.4	44.3	62.8	72.8	79.2	82.6	82.1	75.1	61.1	45.8	33.2	(23)
(24)		5%	MCWB	30.0	31.7	39.5	49.8	56.8	65.4	70.3	68.6	63.2	51.8	40.1	30.6	(24)
(25)		10%	DB	26.9	29.6	39.5	57.4	67.9	75.8	79.7	78.6	70.3	56.3	41.2	29.6	(25)
(26)		10%	MCWB	25.3	27.8	36.3	46.8	54.4	63.9	68.1	66.7	60.4	49.4	37.5	28.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.0	37.4	50.0	57.7	65.7	76.1	78.6	77.0	71.7	60.9	49.7	37.7	(27)
(28)		0.4%	MCDWB	40.6	40.9	58.8	68.9	78.5	84.7	85.7	85.2	79.1	72.9	57.3	41.8	(28)
(29)		2%	WB	33.7	34.4	43.6	53.3	61.8	70.9	74.8	72.8	67.7	56.5	44.2	34.5	(29)
(30)		2%	MCDWB	36.1	36.6	49.9	65.2	70.8	78.1	82.3	82.0	75.9	64.1	49.7	38.0	(30)
(31)		5%	WB	30.1	31.9	39.6	50.3	59.0	67.9	72.3	70.4	64.8	53.0	40.7	31.1	(31)
(32)		5%	MCDWB	32.0	33.9	44.1	60.7	68.6	75.6	79.7	79.1	72.1	59.8	45.3	33.1	(32)
(33)		10%	WB	25.4	27.8	36.5	47.6	56.6	65.4	69.9	68.2	62.0	49.7	37.5	28.2	(33)
(34)		10%	MCDWB	26.8	29.6	39.7	57.3	65.8	73.0	77.7	76.1	69.1	56.0	41.2	29.8	(34)
(35)	Mean Daily Temperature Range		MDBR	15.3	16.0	16.2	19.7	20.6	18.8	19.1	20.8	19.7	16.3	14.1	13.9	(35)
(36)		5% DB	MCDBR	18.4	17.0	20.4	29.1	28.4	24.4	22.7	26.1	27.4	25.6	21.2	17.1	(36)
(37)		5% DB	MCWBR	16.3	14.7	14.4	16.8	14.5	13.0	12.5	13.8	15.7	15.5	14.6	14.5	(37)
(38)		5% WB	MCDBR	18.0	16.8	19.3	26.9	22.7	21.1	21.3	23.2	24.2	24.1	19.9	16.5	(38)
(39)		5% WB	MCWBR	16.2	14.4	14.0	16.4	13.3	13.3	13.2	14.0	15.4	15.5	14.3	14.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.262	0.282	0.299	0.356	0.374	0.378	0.388	0.381	0.347	0.300	0.268	0.248	(40)	
(41)		taud	2.267	2.228	2.342	2.334	2.345	2.364	2.349	2.358	2.445	2.549	2.467	2.347	(41)	
(42)		Ebn at Noon	249	272	289	282	280	279	274	270	270	268	251	241	(42)	
(43)		Edh at Noon	22	30	32	36	37	37	37	35	29	22	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	448	826	1241	1597	1754	1873	1984	1636	1185	711	394	319	(44)	
(45)		RadStd	38	59	97	112	147	131	83	114	90	93	55	36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	-1.01	-1.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page